

Research paper

How to Forecast Non-Obvious Innovations

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2th September 2026

Abstract

Companies forecast innovation with instruments built for forecasting sales: they extrapolate the attribute the market rewards today and call the slope the future. This is pattern recognition, and pattern recognition reads frequency in a remembered series. It is silent exactly where innovation happens — at the jump, when a lighter embodiment dissolves an obligation buyers had stopped noticing, an event absent from any series because it has not yet happened. This paper proposes path recognition as the alternative operation: reading the dependency structure that holds an embodiment in place — its obligations, its identity chain, the capabilities that could change it — rather than the frequency of its past. The paper separates four words the roadmap conflates (forecast, prediction, expectation, desire), places eight established forecasting families on the map of what they read and what they output, and states plainly where a fat-tailed environment breaks each of them, drawing on Kahneman's account of the heuristics that generate pattern recognition and on Taleb's account of the tails that defeat it. It surveys how six other fields already forecast under related conditions — meteorology, epidemiology, seismology, venture capital, military net assessment, and the deliberate culture-building studied in team research — and treats the innovating organisation as a system of stocks, flows and feedback, in the sense used in systems theory, so that an innovation jump becomes a leverage-point intervention rather than a parameter tweak. A specific model is proposed and specified for testing: four measurable variables, a coding protocol, a historical backtest and a prospective trial, each with a stated failure condition. Extreme counterfactuals are then run across five unrelated domains — telephony, books, drinking water, photography and home heating — to stress the model outside its country of origin, alongside three deeper company forecasts (Versuni, Apple, OpenAI), each with a jump, a set of directions to leave alone, a disappearance risk and a falsifier. The paper closes with the positioning dissection a company can run on itself, the forecast machine that runs it quarterly, and the limits of the whole claim. Every forecast produced here is a method-generated hypothesis; the tests, ahead of the argument, decide it.

Keywords: innovation forecasting; pattern recognition; path recognition; fat tails; systems thinking; counterfactual reasoning; disappearance test; heuristics and biases

The paper in one page

The question. For a given company, from its present position, which innovation is most likely to work next, and which directions should it leave alone?

The diagnosis. Every established forecasting family — trend extrapolation, diffusion curves, Delphi panels, tournaments, markets, scenarios, lead-user studies, language-model ensembles — performs pattern recognition: it reads the frequency of a remembered series and projects it forward. An innovation jump is precisely the event missing from that series, because the embodiment that will carry it still awaits existence in the buyer's experience.

The alternative. Path recognition reads structure rather than frequency: the obligations a buyer currently carries, the identity chain attached to the present embodiment, and the dependency changes a given company's capabilities could make. It asks which obligation dissolves next, by which embodiment, from which position, and it is falsifiable by a stated test.

Why the tails matter. Kahneman's programme on judgment shows why people and organisations default to pattern recognition even when it fails: availability, representativeness and the planning fallacy make the familiar pattern feel like knowledge. Taleb's programme on uncertainty shows why the failure is structural rather than a matter of trying harder: innovation lives in a fat-tailed environment (his Extremistan), where a single unrepeated event moves the whole outcome, and a method that fits smooth curves to a smooth past has no purchase there.

The systems view. An innovating company is a system of stocks (its installed base, its trust, its distribution), flows (obligations paid every day by every buyer) and feedback loops, in the sense developed in systems theory. A jump is an intervention at a high-leverage point in that system — a rule, a goal, a structure — rather than at a low-leverage one — a parameter, a number, a feature.

The model. Four variables, coded from public and historical record: obligations removed, identity-chain continuity, capability match, and frontier gap. A historical backtest scores roughly twenty documented technology transitions on all four and tests whether they predict which side won, against a trend-extrapolation baseline. A prospective protocol scores candidate jumps for a live company before the outcome is known.

The stress test. Five unrelated domains — telephony, books, drinking water, photography, home heating — each dissected the same way, plus three deeper company forecasts. The same four variables, the same test, five different histories.

The machine. A positioning dissection run quarterly, a disappearance test on every own product each cycle, three lists — go, leave alone, retire — and one jump with a test card, published yearly with its falsifiers.

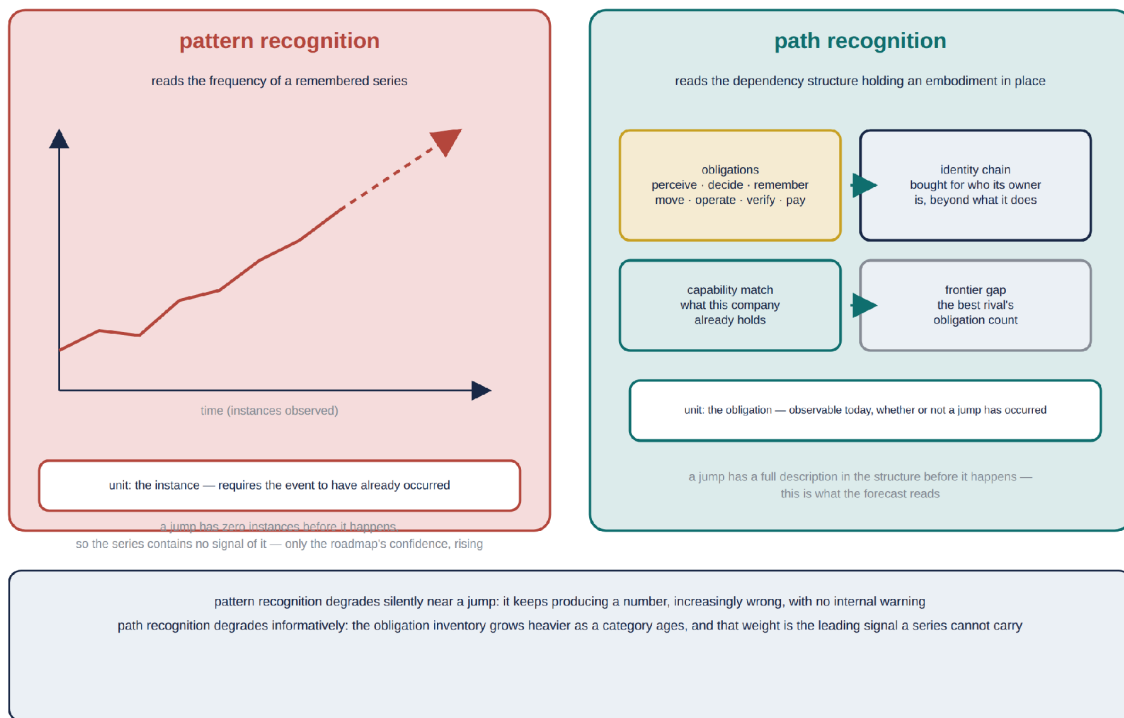
1. Introduction: from pattern to path

An innovation roadmap is a forecast in disguise. When a company writes "next year: a faster, quieter, cheaper version," it has forecast that the attribute the market rewards today keeps being the attribute that matters. That forecast is a pattern read off a series — sales by quarter, ratings by release — and projected forward on the assumption that the series will keep behaving as it has behaved. The forecast is usually right for a year and wrong at the one moment that decides the company's decade: the moment a lighter embodiment removes an obligation buyers had stopped noticing, and the attribute the roadmap was improving stops mattering. The whale-oil candle was at its brightest when the kerosene lamp arrived; the film camera was at its sharpest when the phone camera arrived. Each event is absent from the series that preceded it, being the first instance of its own kind.

This is the general limit of pattern recognition applied to the future: a pattern is a regularity abstracted from a set of past instances, and a jump is, by construction, its own first instance. The companion paper on the Freedom Path Innovation Method [1] dissected this failure at the level of a single product request — the faster-horse problem — and built a method for generating alternatives to a fixed embodiment. This paper dissects the same failure at the level of a forecast, and proposes path recognition as the operation that reads what a pattern leaves out: the dependency structure — obligations, identity, capability — that holds the present embodiment in place, and that a jump changes.

Key points.

- Pattern recognition operates on a series: it asks what has repeated, and projects the repetition forward.
- Path recognition operates on a structure: it asks what a buyer is obligated to do today, and which of those obligations a specific company's capabilities could dissolve.
- A jump has no frequency in any series before it happens; it has a full description in the structure before it happens. The forecast this paper proposes reads the structure.
- This is not a claim that trend methods are wrong. They are the right tool for what becomes passible — a sensor's price, a battery's density. They are the wrong tool for which passible path a buyer actually takes, which is a question about obligations rather than about slopes.

Figure 1 · Pattern recognition and path recognition, contrasted**Figure 1. Pattern recognition and path recognition, contrasted.** What each operation reads, what it outputs, and where each fails.

2. Four words the roadmap conflates

Most roadmap arguments are arguments between people using four different words as if they were one word. Separating them is the first measurement (Table 1; Figure 2).

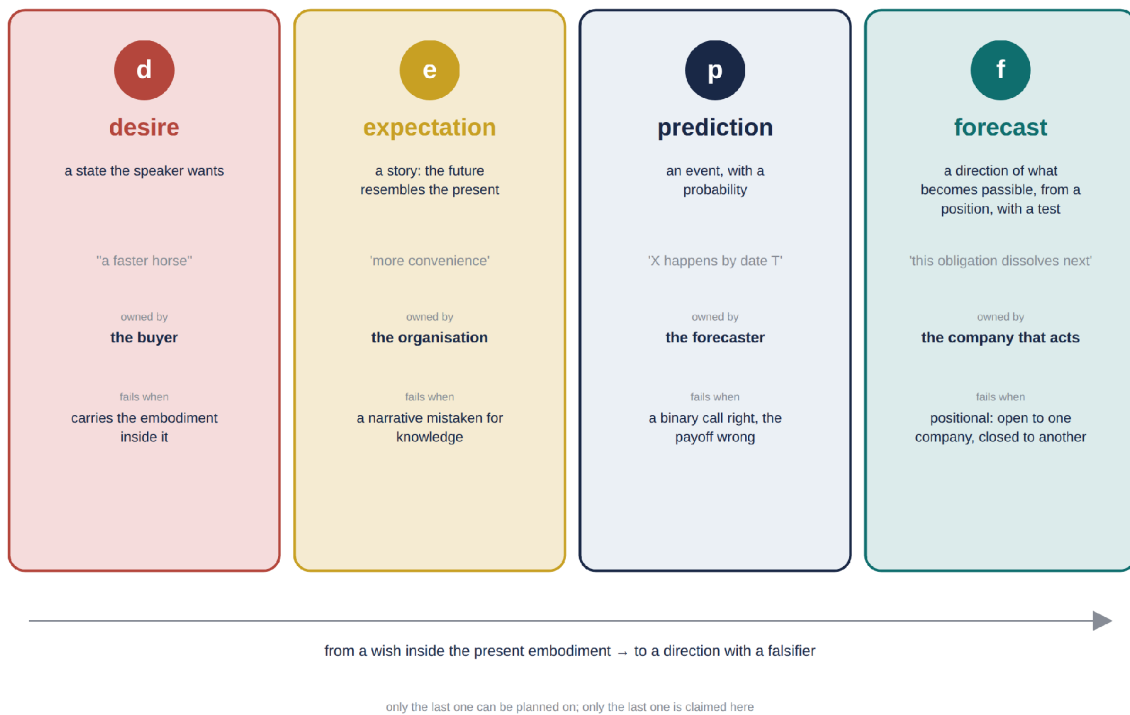
Table 1. Forecast, prediction, expectation, desire.

Word	Its object	Who owns it	Measured against	Characteristic failure
Desire	A state the speaker wants ("a faster horse")	The buyer or the sponsor	Whether the state is reached	Names the embodiment inside the wish; the obligations of the embodiment ride along, unstated
Expectation	A story of how the future resembles the present	The organisation	Whether the story held	The narrative fallacy [10]: a coherent story about the past felt like knowledge of the future
Prediction	A statement that an event occurs, with a probability	The forecaster	Calibration and resolution over many events	Binary call right, payoff wrong [12]: the event was named and the size of its consequence was missed
Forecast (path-recognised)	A statement of direction and structure — what becomes passible, from a stated position, with a test	The company that acts	Whether the direction held and the test passed	Positional error: the jump named is open to a different company than the one reading it

- A desire carries the embodiment inside it; a roadmap built on desires is a list of attribute improvements to that embodiment.
- An expectation is a narrative — "our customers want convenience, so next year's product is a more convenient version of this one" — and it sounds like a forecast while describing the present.
- A prediction answers "will X happen by date T," with a literature of calibration and tournaments [7, 8]. A jump resists that shape: the interesting question is which of many possible events

removes the most obligation, and from whose position, rather than whether one named event occurs.

- A forecast, in the sense used here, states a direction and a structure: which paths become possible, for whom, at what carrying cost. It is the only one of the four a company can plan on.

Figure 2 · Four words the roadmap conflates**Figure 2. Four words the roadmap conflates.** Desire, expectation, prediction and forecast, with the object, the owner and the characteristic failure of each.

3. Why this forecast matters, compared with other forecasts

This section states, in comparative terms, what the path-recognition forecast is for and what each established alternative is for, so the difference is a difference in kind rather than a difference in polish.

3.1 What is at stake

- A trend forecast wrong by a year costs a quarter of misallocated development spending. A jump forecast missed costs the category: the company that keeps improving the candle while a rival ships the bulb loses the market for light, not the market for candles.
- The cost of a missed jump is asymmetric and delayed. It shows up as a healthy income statement the year before the category turns, which is the reason boards approve one more year of the faster-horse roadmap: every number they can see still points the old way.
- The forecast proposed here is built to be read for stakes rather than for elegance: each company card in Section 9 states a disappearance risk — a named product of the company's own that the jump, if real, deletes. A forecast that leaves the forecaster's own portfolio untouched has looked in the wrong place.

3.2 Eight established families, measured against this one

Table 2 measures eight established forecasting families on the same six questions, and adds what each is for and what it necessarily misses, so the comparison states purpose alongside mechanism.

Table 2. Eight established forecasting families, compared with path recognition.

Method	What it is for	Reads	Outputs	Sees a jump?	Where it fails
Trend extrapolation	Near-term volume planning	The past of an attribute	A slope with an interval	Blind: a jump is a break in the series it fits	Highest confidence appears immediately before the break [10]

Method	What it is for	Reads	Outputs	Sees a jump?	Where it fails
Diffusion and S-curves (Bass, Rogers)	Pace of adoption, once named	Adoption of a named embodiment	An adoption curve	Blind: the embodiment must already be named as an input [2, 3]	The thing to be forecast enters as a premise
Delphi panels	Aggregating expert judgment	Expert belief, iterated to consensus [5]	A consensus statement	Weak: experts reason inside the current embodiment	Consensus compresses exactly the outlying view that saw the jump
Superforecasting tournaments	Calibrated probability on resolvable questions	Many scored questions	Calibrated probabilities [7, 8]	Only if the jump is already phrased as a question	The question must exist before the jump is visible
Prediction markets	Aggregating dispersed information into a price	Prices on resolvable propositions	A price	Same limit as tournaments	Fat-tailed payoffs are mispriced by design [11]
Scenario planning	Rehearsing several futures at once	Driving forces, combined	Coherent narratives	Sometimes, if a scenario happens to contain it	Judged by coherence rather than by a test; an expectation in costume
Lead users and jobs-to-be-done	Surfacing an unmet outcome	What advanced users already do; stated outcomes	A need statement	Partly: sees the need, misses the embodiment that removes it	Users describe outcomes in the vocabulary of the embodiment they already hold [1]
Language-model ensembles	Fast synthesis of existing text	Text of the present	Text	Weak: measurably less diverse than human ideation	Fixation is now measured inside the models themselves [1, 22]
Path recognition (this paper)	Naming the jump, its embodiment, and who can make it	Obligations, identity chain, capability, frontier	A direction with a test	Yes, by construction: the object read is the structure a jump changes	Ordinal rather than dated; positional rather than universal — Section 11

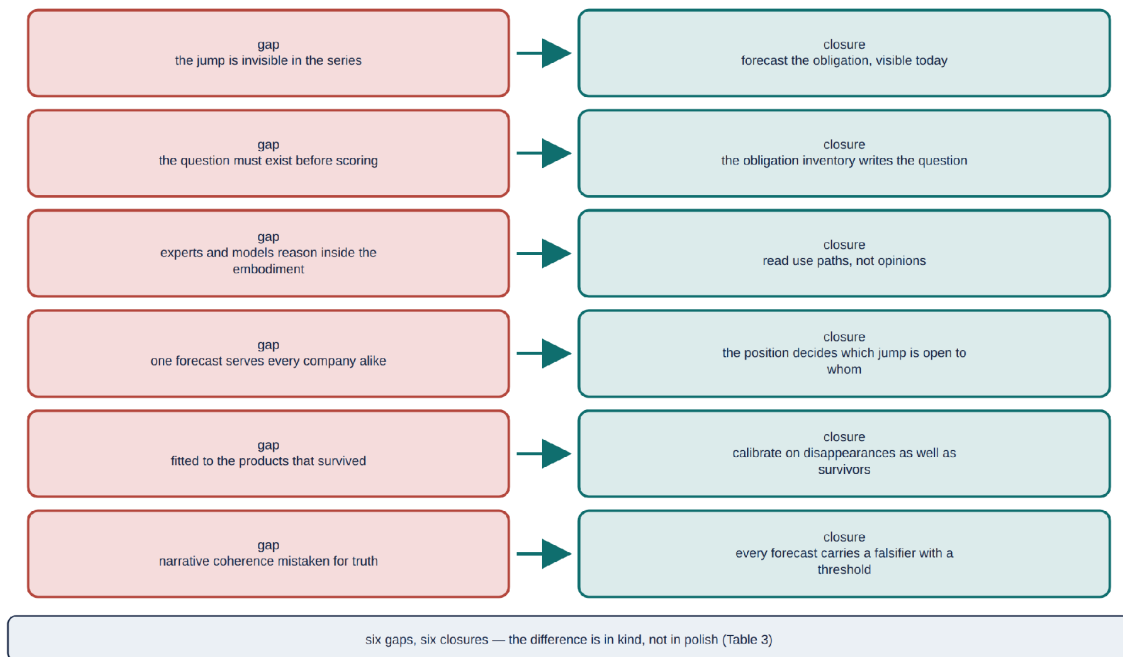
Key points.

- The first eight rows share one property: each reads either the past of an embodiment or the stated beliefs of people who live inside it. Each reads something else entirely, leaving the obligation ledger unexamined.
- The families with the best calibration record — tournaments, markets — require the question to be phrased before scoring begins. Phrasing the jump as a question is the entire difficulty; path recognition supplies the phrasing mechanically, from the obligation inventory, rather than waiting for an analyst to think of it.
- Kahneman and Lovallo's account of the roadmap is exact: organisations are bold about the slope of the present and timid about leaving it [6] — timid choices, bold forecasts, precisely reversed from what a jump requires.
- The comparison is not a ranking. Trend methods remain the correct tool for what becomes passible; path recognition is the correct tool for which passible path is taken, and for by whom.

3.3 What path recognition closes, stated as gap and closure

Table 3. Gaps left open by the eight established families, and how path recognition closes each.

Gap	Left open by	Closure
The jump is invisible in the series	Trends, S-curves, markets	Forecast the obligation, which is visible today, rather than the sales of the embodiment that will remove it
The question must exist before scoring can start	Tournaments, markets, Delphi	The obligation inventory generates the question; the disappearance test (Section 6) phrases it
Experts and models reason inside the present embodiment	Delphi, lead users, language models	Read the obligation inventory from use paths rather than from opinions; opinions enter only as frontier facts
One forecast serves every company alike	All eight	The position — capabilities, identity chain, distribution — decides which jump is open to which company
Fitted to the products that survived	Nearly all	Calibrate on products that disappeared as well as on survivors that kept an identity chain (Section 6)
Narrative coherence mistaken for truth	Scenarios, expectations	Every forecast this method produces carries a falsifier with a threshold and a duration

Figure 3 · Gaps left open by eight established families, and their closure**Figure 3. The gaps left open by eight established forecasting families, and their closure.** Read alongside Table 3.

4. Pattern recognition versus path recognition

4.1 Why organisations default to pattern

Kahneman's account of judgment under uncertainty gives the mechanism [25, 26, 27]. Three heuristics, each efficient and each a source of the roadmap's blind spot:

- **Availability.** An event is judged more likely the more easily instances come to mind. The faster horse comes to mind easily, because horses are what the buyer has ridden; the car stays absent from mind, since riding one is still a first experience. Availability makes the embodiment the buyer already owns feel like the whole space of possible answers.
- **Representativeness.** A new case is judged by how well it resembles a familiar category, independent of the base rate. A device with a screen is judged as a phone, a chat window is judged as a product category unto itself, and both judgments smuggle in every obligation the category already carries.
- **The planning fallacy.** Plans are drawn from an inside view — this project, our team, our roadmap — that ignores the outside view of how such plans have gone before; forecasts are bold about the plan's own trajectory and timid about anything that would depart from it [6]. A roadmap is an inside-view document by construction.

Each heuristic is a shortcut for pattern recognition, and each shortcut is precisely wrong at the jump, because the jump is defined by an absent prior instance, an absent established category, and an absent place inside the current plan.

4.2 Pattern recognition and path recognition, contrasted

Table 4 (continuing the numbering as Table 4 below is reserved for tail behaviour; see Table 6). *(kept as prose contrast, not a table, for density.)*

- Pattern recognition asks: what has this series done before, and how much of that? It answers with a number and an interval.

- Path recognition asks: what must a buyer do today to reach the desired state, and which of those obligations can a given company's capabilities remove? It answers with a direction and a test.
- Pattern recognition is backward-referential: its unit is the instance, and an instance requires the event to have already occurred at least once.
- Path recognition is structure-referential: its unit is the obligation, and an obligation is observable in the present, whether a jump has occurred to remove it or is still pending.
- Pattern recognition degrades gracefully near a jump: it keeps producing a number, increasingly wrong, carrying its own confidence as the only signal, right up to the moment that confidence turns out to have been misplaced.
- Path recognition degrades informatively near a jump: the obligation inventory grows heavier as a category ages (more consumables, more steps, more exceptions bolted onto the original mechanism), and that accumulation is itself the leading indicator a pattern-based method cannot represent.

[[FIG 1 is placed in Section 1; this contrast is carried forward into Figure 4's fourth-quadrant map.]]

5. Fat tails: forecasting under Extremistan

5.1 Two kinds of randomness

Taleb's distinction between Mediocristan and Extremistan gives the reason pattern recognition is more than inconvenient near a jump; it is structurally mismatched there [10, 11]. In Mediocristan, any single observation moves the aggregate by very little — human height, calorie intake, a fair die — and the tools of classical statistics, means, variances, confidence intervals, work as advertised. In Extremistan, a single observation can dominate every other observation combined — wealth, book sales, and the outcome of an innovation category — and those same tools become a description of the past with limited claim on the future.

- Category revenue before a jump looks like Mediocristan: many small transactions, a stable mean, a narrow variance.
- Category revenue after a jump looks like Extremistan: one embodiment change accounts for nearly all the movement, and the pre-jump variance said nothing about its size.
- A forecasting method built for Mediocristan — trend extrapolation, most diffusion modelling, most scenario weighting — is a Mediocristan tool applied to an Extremistan question, and the mismatch is invisible until the tail event arrives.

5.2 The fourth quadrant, populated

Taleb sorts decisions into four quadrants by two properties: whether the payoff is simple (binary, bounded) or complex (open-ended, unbounded), and whether the underlying distribution is thin-tailed or fat-tailed [11]. The first three quadrants are where standard forecasting tools are safe to use. The fourth — complex payoffs under fat tails — is where they fail, and it is exactly where an innovation jump lives: the payoff (a category won or lost) is unbounded, and the distribution (how large the winning embodiment's share becomes) is fat-tailed.

Figure 4 · The fourth quadrant, populated

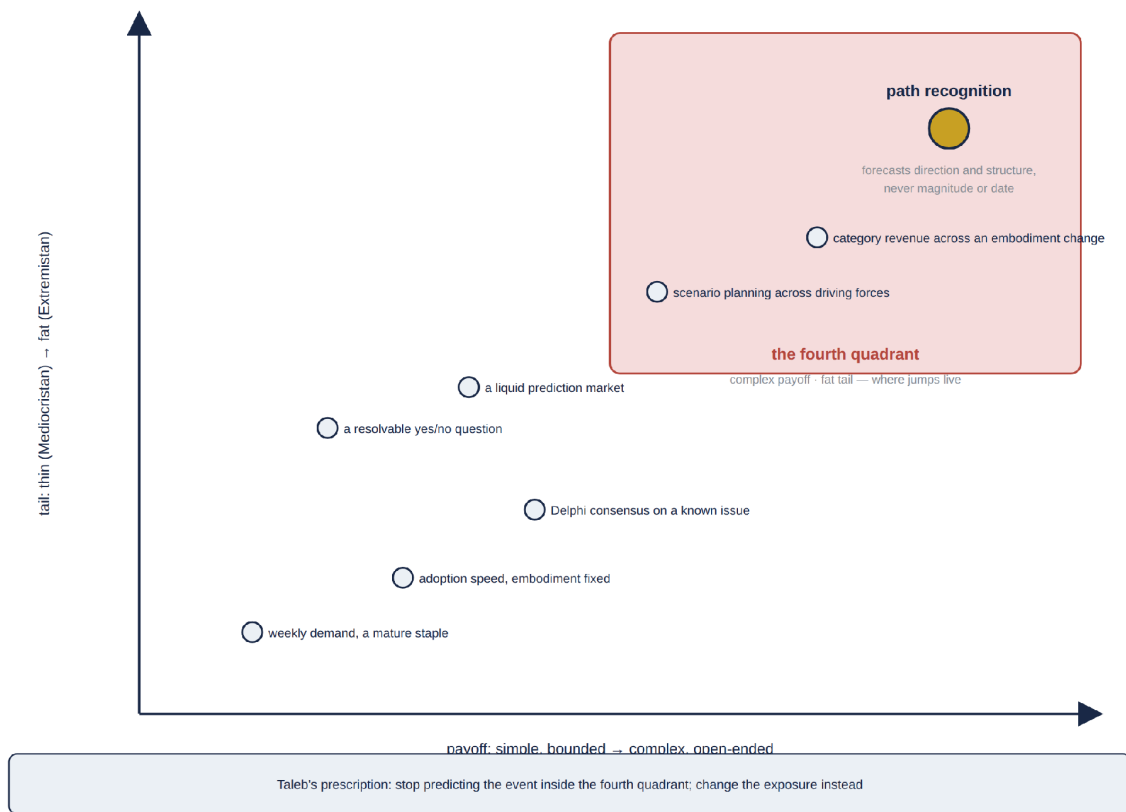


Figure 4. The fourth quadrant, populated. Thin-tailed and fat-tailed forecasting problems, with the established families placed by payoff complexity and tail behaviour.

Table 5. Where established forecasting tools sit relative to the fourth quadrant.

Property	Thin-tailed, simple payoff (safe)	Fat-tailed or complex payoff (fourth quadrant)
Trend extrapolation	Weekly demand for a mature staple	Category revenue across an embodiment change
Diffusion curves	Adoption speed once the embodiment is fixed	Which embodiment gets adopted
Tournaments and markets	A resolvable yes/no question with a bounded prize	An open-ended "what wins the category" question
Path recognition	—	Designed for this quadrant: it forecasts direction and structure rather than magnitude or date

5.3 What Taleb's programme prescribes, and how the forecast follows it

- **Via negativa.** The future is forecast more reliably by what will be removed than by what will be added [14]. Path recognition is subtractive by construction: its unit of measurement is the obligation removed, never a feature added.
- **The narrative fallacy.** A coherent story compresses many possible pasts into one, and the compression is mistaken for insight [10]. The obligation inventory resists this by staying a ledger — items listed rather than a story told — so that the count, rather than a single narrative, carries the claim.
- **The turkey problem.** A confidence built from a thousand days of consistent feeding reaches its peak the day before the feeding stops [10]. A category's trend forecast reaches its peak

confidence immediately before a jump, because every quiet day adds one more confirming observation to a series about to become irrelevant.

- **Skin in the game.** A forecast that carries zero cost when wrong invites careless issuance [21]. Every jump in this paper carries a falsifier with a population, a duration and a threshold (Section 9), so that being wrong has a stated, checkable cost to the claim.
- **Antifragility.** A portfolio gains from disorder when its exposure to a favourable tail is large and its exposure to an unfavourable one is capped [14]. The disappearance test (Section 6) caps a company's unfavourable exposure by naming its own products most likely to be deleted, before a rival deletes them.

Key points.

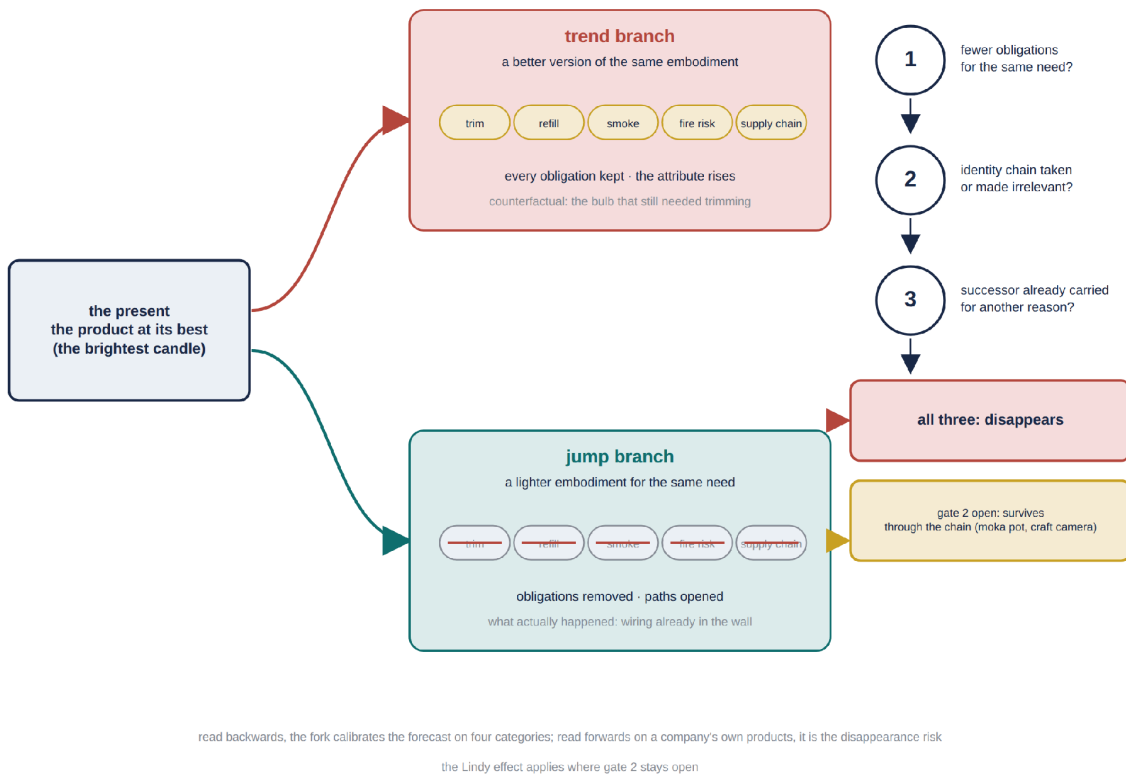
- The fourth quadrant is not a rhetorical flourish; it is the reason the eight established families in Table 2 stay unrepaired by more data or a longer window. Extra data from a thin-tailed regime describes only that regime, leaving a fat-tailed one untouched.
- Path recognition does not attempt to forecast the magnitude or the date of the tail event. It forecasts its direction and its structure, which are the two properties that survive fat tails, because they are read from what already exists rather than extrapolated from what has already happened.

6. Counterfactuals: how the forecast is tested against the past

6.1 The disappearance test

A forecast of jumps can be tested backwards. For a product that disappeared, ask the counterfactual: had the successor carried the same obligations, would the product have survived? For a product that survived, ask which condition stayed open. Three conditions, a test a product must fear together:

- Another embodiment serves the primary need with fewer obligations.
- That embodiment takes the identity chain, or makes it irrelevant.
- That embodiment is already carried or owned for another reason, so its carrying cost is near zero.

Figure 5 · The counterfactual fork and the three gates of disappearance**Figure 5. The counterfactual fork and the three gates of the disappearance test.** From the same present, the trend branch keeps every obligation and the jump branch removes them.

Read through the fork of Figure 5:

- **Whale-oil candle.** Counterfactual: a bulb that still needed trimming and refilling would have left the spermaceti trade in place. The bulb removed trimming, smoke, refilling and fire risk (one), made light unremarkable and emptied the candle's identity chain (two), and rode on wiring installed for other reasons (three). The candle survives where the identity chain is the product: ceremony and decoration.
- **Film and the stand-alone camera.** Counterfactual: a phone camera that required a cable to move an image would have left the camera its place. The phone removed carrying a second object, transferring and deciding to bring it, took the identity chain, and was already in the pocket. The craft camera survives, because identity and capability sit in the camera itself for a craft photographer.
- **The moka pot.** Capsule and bean-to-cup machines removed grinding, timing, watching and cleaning (one) and were bought for the kitchen anyway (three). Condition two held only in part: the moka pot kept a ritual chain, and retreated rather than vanished — the case that separates the test from a plain obligation count.
- **The slow oven.** The microwave and the air fryer removed pre-heating, waiting and oil for meals that fit a small cavity. Condition one held only for a subset of meals, which is why the forecast retires a use rather than a product.

6.2 A documented case where the jump was invented first and adopted last

The clearest illustration of path recognition failing to be acted on, even after the structure was fully known internally, is the digital camera at Kodak. Engineer Steven Sasson built a working digital

camera inside Kodak's own laboratories in 1975 [35]; the obligation-removal (film, processing, waiting) and the capability match (the company's own optics and its own distribution) were both present inside the same organisation that would, decades later, be dissolved by exactly that jump made by rivals. The historical record is more layered than the popular telling suggests — Kodak continued developing the technology internally and licensed early sensors rather than suppressing the idea outright — but the structural point survives the correction: capability and knowledge of the obligation ledger were present, and the company's own identity chain (the film brand, the processing business) supplied the resistance, ahead of any shortage of foresight. Extreme counterfactual: had Kodak treated its 1975 prototype as a positioning dissection input rather than as a threat to an existing revenue line, the jump it eventually lost to Canon, Sony and the smartphone camera could have been the jump it made on its own terms — the outcome the forecast machine in Section 10 is built to produce.

Key points.

- The Lindy effect states that the expected remaining life of a non-perishable technology grows with its age [14]. The disappearance test states when Lindy applies: to objects with an identity or ritual chain, and to needs whose obligations are already minimal. An object that is only a bundle of obligations is fragile whatever its age.
- The Kodak case shows the test applies inside a company as well as across a market: capability and knowledge of the jump can sit unused when an identity chain built on the old embodiment supplies the resistance.

7. How other fields already forecast under related conditions

Innovation forecasting is not the first field to face a fat-tailed, structure-dependent future. Six fields already forecast under related conditions, each with a named real-world instrument, compared for what transfers (Table 6).

Table 6. How six fields predict under conditions related to an innovation jump.

Field	What it forecasts	Method, with a real instrument	What transfers to innovation forecasting
Meteorology	Weather several days out	Ensemble forecasting: many physical simulations from slightly different starting conditions, combined into a probability spread — the operational method behind the European Centre for Medium-Range Weather Forecasts' global system [36]	Report a spread of jumps and their obligations in place of a single point forecast; widen the spread as the horizon lengthens
Epidemiology	The course of an outbreak	Compartmental models (susceptible–infected–recovered) driven by a reproduction number rather than by extrapolating case counts, following Kermack and McKendrick's founding formulation [37]	Track the reproduction rate of an obligation's removal — how fast the new embodiment's adopters convert others — rather than the raw adoption count
Seismology	Earthquake hazard, not earthquake events	Explicitly refuses point prediction of timing; forecasts hazard as a probability over a region and a period, from accumulated fault stress — the method behind the US Geological Survey's Uniform California Earthquake Rupture Forecast [38]	Report where the stress (the obligation) has accumulated and how much, rather than claim a date for its release
Venture capital	Which of many founders builds a durable company	A stated preference for "definite" theses — a specific, falsifiable claim about which monopoly-scale jump a founder is making — over pattern-matching against prior winners [39]	Prefer a definite, falsifiable jump statement over a pattern match to the last successful category; check every claim against a base rate
Military net assessment	An adversary's capability and intent, ahead of a date of conflict	Comparing capability, doctrine and position rather than counting past incidents, the discipline built inside the Pentagon's Office of Net Assessment under Andrew Marshall over four decades [40]	Compare capability and position (Section 12's basis point) rather than an opponent's — or a rival's — past product cadence

Field	What it forecasts	Method, with a real instrument	What transfers to innovation forecasting
Team and organisational culture	Whether a group will surface a non-obvious signal rather than default to the obvious one	Deliberate construction of psychological safety [33], documented in practice at Google's Project Aristotle team-effectiveness study, Pixar's Braintrust review process, and the San Antonio Spurs, three cases examined together in Coyle's account of high-performing groups [32]	An organisation's capacity to run path recognition depends on whether dissent from the roadmap's pattern is rewarded or punished internally, before it depends on any external method

Key points.

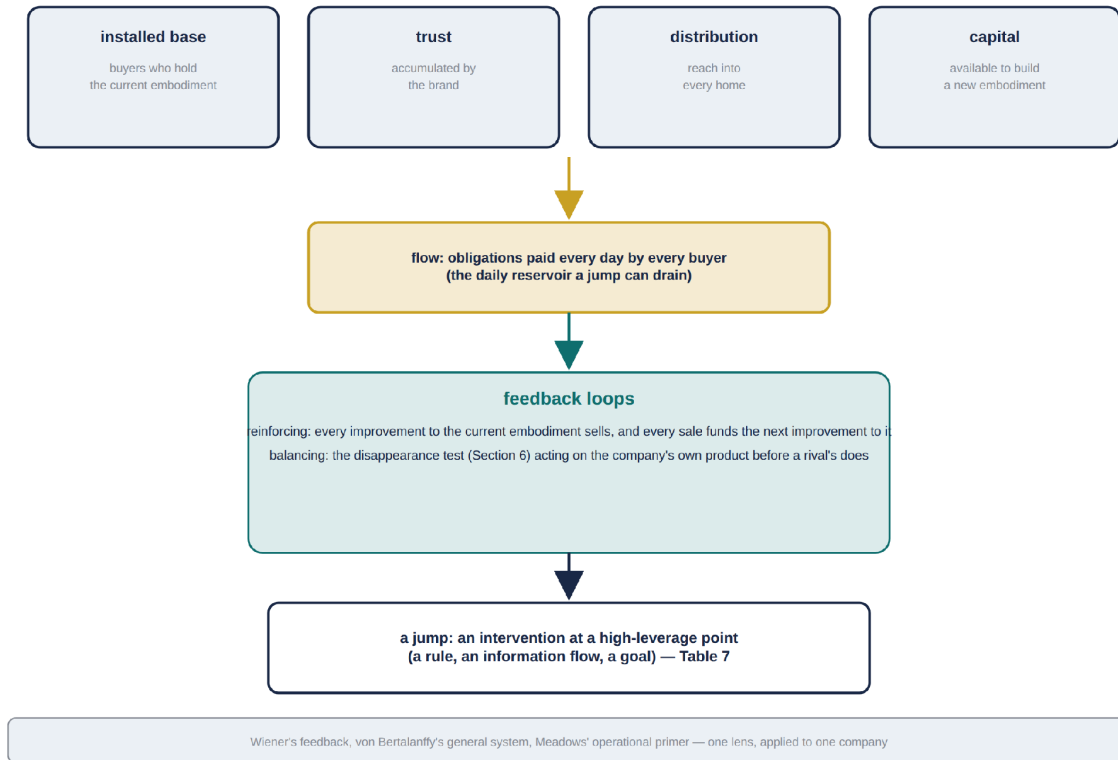
- Every field in Table 6 that forecasts well under uncertainty has abandoned the ambition of a single-point prediction.
- Each replaces the point prediction with a structural or distributional statement instead — a spread, a hazard, a capability comparison, a definite thesis, a culture.
- Path recognition follows the same discipline: it states a direction and a test, never a date and a magnitude.
- The culture row matters for a reason specific to organisations rather than to markets: Google's Project Aristotle found psychological safety to be the strongest predictor of team effectiveness among the variables it tested, ahead of individual talent or tenure; Pixar's Braintrust and the Spurs' locker-room culture are the qualitative cases built around the same finding [32].
- Edmondson's original construct — a team climate in which taking an interpersonal risk feels safe — is the measured basis for all three cases [33].
- An obligation inventory (Section 8) is only as complete as the willingness of the person closest to the buyer to name an obligation the roadmap has stopped noticing.

8. The universe as a systems machine

8.1 Stocks, flows and feedback

Treating a company, a market, or a household as a system — a set of stocks connected by flows, regulated by feedback loops — is the founding move of systems theory, from Wiener's early formalisation of feedback and control [30] through von Bertalanffy's general account of systems that hold across biological, mechanical and social domains [31] to Meadows' operational statement of the same ideas for practitioners [29]. The move applies cleanly to an innovating company:

- **Stocks.** The installed base of buyers who already hold the current embodiment; the trust a brand has accumulated; the distribution network reaching those buyers; the capital available to build a new embodiment.
- **Flows.** The obligations paid every day by every buyer of the current embodiment — the daily reservoir a jump can drain. Also, the rate at which a new embodiment converts holders of the old one.
- **Feedback.** A reinforcing loop keeps the roadmap improving the current embodiment, because every improvement sells, and every sale funds the next improvement to the same embodiment. A balancing loop — when it exists — is the disappearance risk acting on the company's own product before a rival's does.

Figure 6 · The company as a system: stocks, flows and feedback**Figure 6. The company as a system: stocks, flows and feedback.** Applied to an innovating company's installed base, trust, distribution and obligation flow.

8.2 Leverage points, and why a jump is a high-leverage intervention

Meadows ranks the places to intervene in a system from least to most powerful: adjusting a parameter or a number is the weakest lever; changing the rules, the information flows that feed a decision, or the goal the system is pursuing are far stronger [29]. The faster-horse roadmap is, in this language, an intervention at the weakest leverage point available — a parameter of the existing structure (speed, size, price) — while a jump is an intervention at a structural leverage point: a rule (what the product is obligated to do), an information flow (what the buyer must perceive to use it), or the goal itself (arrival, rather than horse-keeping).

Table 7. Meadows' leverage points, applied to the faster-horse roadmap and to a jump.

Leverage point (weak → strong)	The faster-horse move	The jump
Parameters, numbers	Faster, cheaper, quieter	—
Buffers and stocks	A bigger fuel tank, a bigger battery	—
Structure of material flows	A better factory line for the same product	A new supply chain for a removed consumable
Feedback loop strength	Louder marketing of the same attribute	A loop that removes the obligation to notice at all (the deadline cooker, Section 9)
Information flows: who sees what	A dashboard showing the same reading	The signal reaching the actuator directly, so no one has to look
Rules: what the system is obligated to do	An added feature, a new setting	A dependency changed: presence, scheduling, or verification moves off the buyer's job
Goals: what the system is for	— (goals are rarely revisited on a roadmap)	Transport rather than horse-keeping; an answer held rather than an answer given

Key points.

- Read as a leverage-point ladder, the faster-horse problem is simply the observation that most roadmaps stay parked on the first two rungs.
- A jump forecast, in this frame, is a claim about which rung a given company's capabilities can reach, rather than about how much better the current rung can be made.
- This framing also explains why a jump so often looks obvious in hindsight and invisible in advance: a higher leverage point changes fewer variables (a rule, a goal) while producing a larger effect, which is exactly the signature Meadows describes as counter-intuitive to people trained to look for effects proportional to effort [29].

9. Path recognition: the model, and how to test it

9.1 Four variables

Table 8. The path-recognition model: variables, definitions and coding rules.

Variable	Definition	Coding rule
Obligation-removal index (ORI)	Count of obligations (perceive, decide, remember, move, operate, verify, clean, maintain, pay) removed by the candidate embodiment relative to the incumbent, for the same primary need	Two independent coders list obligations from the use path of each embodiment; the index is the count removed, agreement checked by Cohen's kappa
Identity-chain continuity (ICC)	Whether the candidate embodiment carries, takes, or leaves empty the identity chain of the incumbent	Coded ordinally: takes it (2), makes it irrelevant (1), leaves it fully intact in a niche (0)
Capability match (CM)	Whether the company evaluated already holds the sensing, actuation, distribution, trust or identity capability the candidate embodiment requires	Coded from public record: patents, existing product lines, distribution agreements, prior statements of capability
Frontier gap (FG)	The obligation-removal index of the best current rival or substitute, subtracted from the candidate's own index	Requires an independent obligation count on the frontier candidate, using the same coding rule as ORI

9.2 The hypothesis

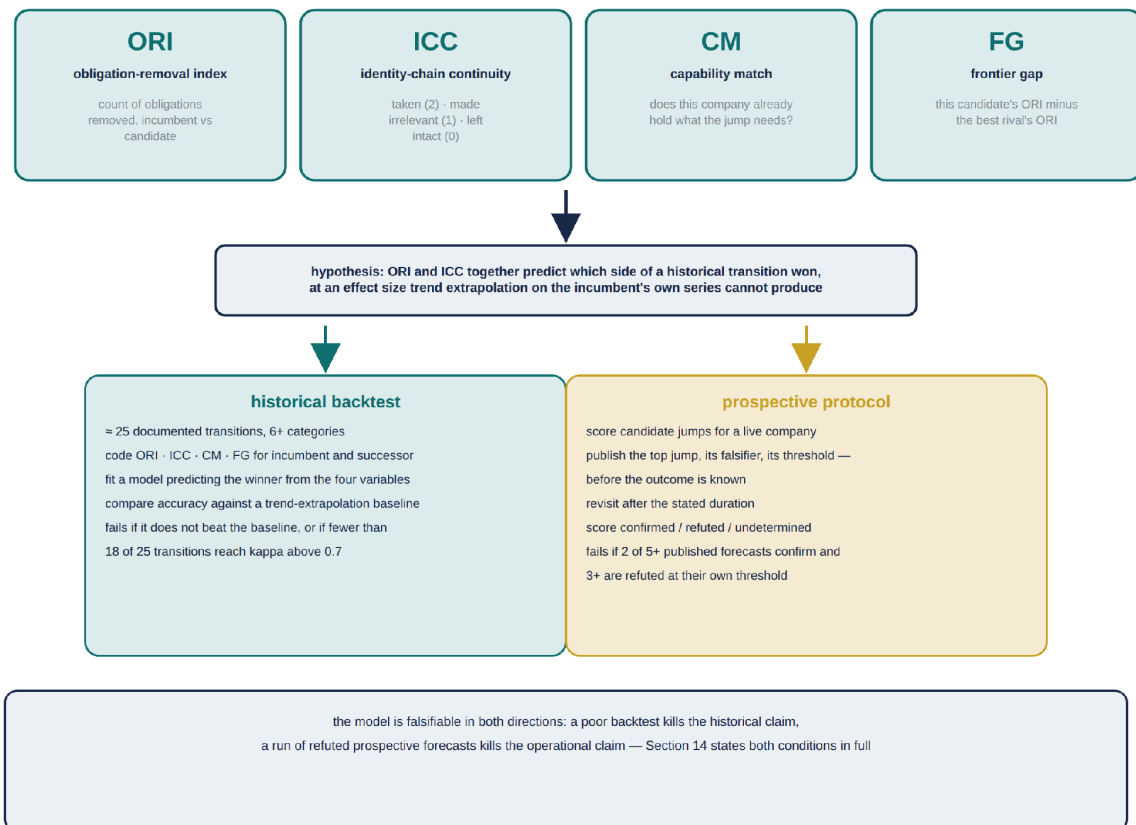
Path recognition predicts that, across documented technology transitions, the incumbent that lost the category is the one for which a successor existed with a higher obligation-removal index and a positive identity-chain-continuity score, while trend extrapolation — fit to the incumbent's own sales series alone — predicts a flat continuation instead. Stated as a test: ORI and ICC, taken together, predict which side of a transition wins, at an effect size trend extrapolation on the incumbent's series leaves unreached, because the successor is absent from the incumbent's series entirely.

9.3 The historical backtest

- Assemble a set of approximately twenty-five documented technology transitions spanning at least six categories (illumination, food preparation, photography, telephony, written media, personal computing).
- For each, code ORI, ICC, CM and FG for the incumbent and for the embodiment that in fact prevailed, blind to the historical outcome where the coder's knowledge allows blinding (a limit acknowledged in Section 11).
- Fit a model predicting the historical winner from ORI, ICC, CM and FG alone, and compare its classification accuracy against a trend-extrapolation baseline that uses only the incumbent's pre-transition sales trajectory.
- Pre-registered failure condition: the paper's claim stands only if the four-variable model out-performs the trend baseline at conventional significance, with at least eighteen of the twenty-five transitions coded to inter-rater agreement above kappa 0.7.

9.4 The prospective protocol

- For a live company, run the positioning dissection (Section 10) to produce a ranked list of candidate jumps, each scored on ORI, ICC, CM and FG.
- Publish the top-ranked jump, its falsifier, and its threshold before the outcome is known — the discipline Taleb calls skin in the game applied to a forecasting method [21].
- Revisit after the stated duration; score the jump as confirmed, refuted, or undetermined against its own threshold.
- Failure condition: the method fails if, across five or more published prospective forecasts, confirmations stay under two while refutations reach three or more at their stated threshold.

Figure 7 · Path recognition: the model and its test protocol**Figure 7. The path-recognition model and its test protocol.** Four variables, a historical backtest, a prospective protocol, each with a stated failure condition.

9.5 Reproducibility

- Coding sheets, the obligation-use-path template, and the twenty-five-transition candidate list are to be published alongside the historical backtest, so an independent team can re-code the same transitions.
- Every prospective forecast is dated and time-stamped at publication, before its threshold window closes, following the pre-registration discipline standard in the forecasting-tournament literature [7, 8].
- Inter-rater agreement (Cohen's kappa) is reported for every coded variable, alongside the pass-or-fail call against the 0.7 threshold, rather than folded into it.
- Coders are trained on a shared rubric before touching the twenty-five-transition set, and a held-out subset is double-coded by a third, independent rater as a check on rubric drift.
- Every company card in Sections 10.1 and 11 states its falsifier in the same paragraph as its jump, so the claim and its refutation condition stay within one page of each other.

10. Extreme counterfactuals across five domains

The disappearance test and the four-variable model are stress-tested here outside the appliance and coffee cases already familiar from the companion paper, across five domains chosen for their distance from one another: telephony, books, drinking water, photography, home heating.

Table 9. Five domains, dissected, with an extreme counterfactual for each.

Domain	Incumbent embodiment	Obligations carried	Successor	Obligations removed	Extreme counterfactual
Telephony	The shared landline, tethered to one location	Waiting one's turn, being reachable only at that location, a party line's absent privacy	The mobile handset, then the smartphone	Location, turn-taking, and eventually the handset itself as a distinct object once every function merges into one	Had battery density stalled at 1990s levels, the mobile phone would have remained a car-mounted device, and the smartphone jump would have waited on a battery ahead of a screen — the direction would have been the same and the date entirely different, which is the paper's claim about what trend methods can and cannot supply
Books	The bound printed volume	Physical storage, carrying weight, the eyes fixed on a page, a single copy in a single place	The audiobook, delivered as a stream	Weight, storage, and the obligation to look at anything at all, in exchange for a new obligation (attention to sound) that many daily activities already leave free	Had the printing press been suppressed by the guilds that opposed it, as was in fact attempted, the identity chain of the hand-copied manuscript would have persisted longer, and the jump to mechanical printing — itself a jump, in its own century — would have happened elsewhere first; the ebook, in this frame, removed only the weight obligation while keeping the look-at-a-page obligation, which is why the audiobook and the podcast, ahead of the ebook, carry the larger share of the obligation actually removed
Drinking water	The communal well or river, requiring transport and a purity risk	Fetching, carrying, uncertainty about safety	Piped and treated water, then bottled and filtered water	Fetching and carrying entirely; purity uncertainty; later, taste and portability	Had municipal treatment stalled short of reliability, bottled water would have inherited the fetching obligation rather than removing a taste preference; the actual sequence — pipe first, bottle second — shows the model's obligation-removal ranking rather than an assumption about which embodiment "should" have come first
Photography	The stand-alone camera and film development	Carrying a second object, buying and developing film, waiting to see the result	The smartphone camera	Carrying the object, film purchase, developing, and the wait; also, transfer to share the image	Kodak's own 1975 prototype (Section 6.2) shows the obligation-removal and the capability match were both available inside the incumbent; the counterfactual sits in "what a positioning dissection, run inside Kodak in 1975, would have recommended against the company's own identity chain"
Home heating	The manually adjusted furnace dial	Remembering to adjust it, guessing at comfort versus cost, the pattern going unrecorded	The learning, connected thermostat	The memory obligation and the guesswork, replaced by a device that infers the pattern and acts on it	Had wireless sensing and cheap microcontrollers arrived thirty years earlier, the same jump would have been available to the furnace manufacturers themselves rather than to a new entrant; the jump was open by capability to incumbents and taken by an outsider, which the positioning dissection's capability-match variable is built to flag before the fact rather than after it

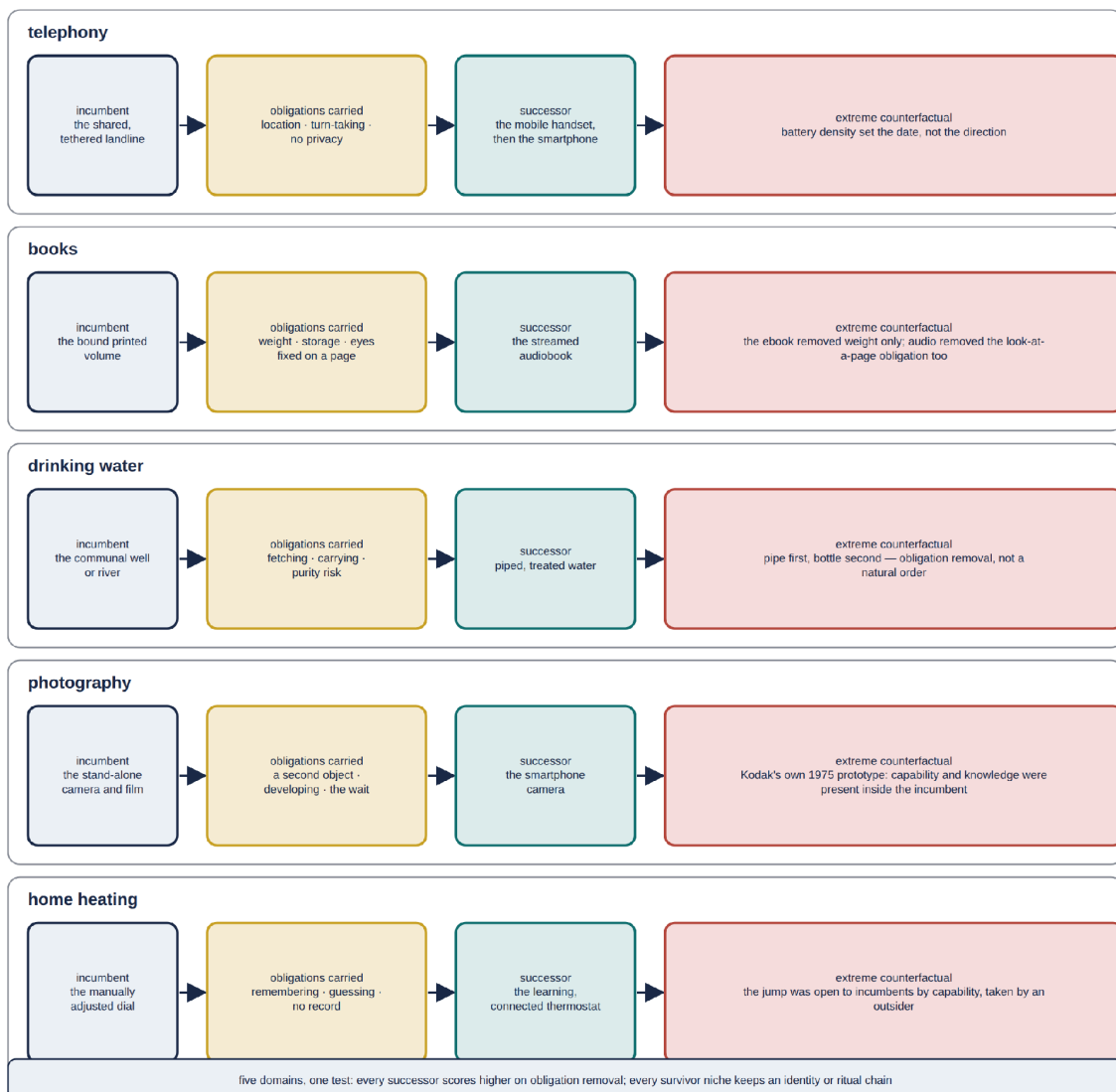
Figure 8 · Five extreme counterfactuals

Figure 8. Five extreme counterfactuals. Telephony, books, drinking water, photography and home heating, each dissected for obligations removed and an alternate branch.

Key points.

- Across five unrelated domains, every successor embodiment scores higher on obligation removal than the incumbent it replaced, and every survivor niche (the manuscript's calligraphy tradition, the printed collector's edition, the bottled mineral water sold on origin rather than on purity, the darkroom print, the manual thermostat kept for its dial as a design object) persists through an identity or ritual chain rather than through a lower obligation count.
- The telephony and heating counterfactuals isolate date from direction: a trend input (battery density, sensor cost) changed when the jump happened; the direction of the jump stayed fixed regardless. This is the paper's central claim about the division of labour between trend methods and path recognition, made concrete outside the appliance domain.

10.1 Method-generated innovation candidates: what is expected next

The disappearance test in Section 6 reads backwards, from an embodiment already replaced. Run forward, the same four variables (Section 9.1) generate candidate jumps for a domain still awaiting its

turn — the operation the method is for. Every candidate below is scored qualitatively on the four variables and grounded in a real, already-shipping precedent, so that "what is expected" means a documented direction already visible at small scale, read forward, rather than an invented scenario. Photography receives the deepest treatment, at the user's request; the remaining four domains follow at the same level of rigour, more briefly.

Photography: three candidates

- **Candidate 1 — passive, cue-triggered capture.**

- Obligation removed: framing, aiming, and deciding when to take the picture — the obligation the smartphone camera left fully in place.
- Real precedent: the Narrative Clip (2013) and Google Clips (2017) both shipped always-on or intermittently-triggered wearable cameras; Ray-Ban Meta smart glasses (2023 onward) ship a camera whose capture skips the phone-retrieval step entirely.
- Identity-chain continuity: takes the "decisive moment" identity chain away from manual composition skill, and relocates it to curation after the fact.
- Capability match: open to companies that already hold a wearable, always-worn device — eyewear and earwear makers, ahead of camera makers.
- Ruled out: continuous, unfiltered recording, which trades the framing obligation for a new and larger one — reviewing everything captured, and the social cost of being recorded by a device that runs continuously.
- What is expected: capture triggered by a cue (a voice cue, a gesture, or a biometric spike) rather than continuous recording, which keeps the obligation-removal gain while the review obligation stays absent.
- Falsifier: 100 users, 30 days; failure if the average time spent reviewing captured material exceeds the time the old camera-retrieval step took.

- **Candidate 2 — provenance built into capture.**

- Obligation removed: proving, after the fact, that an image is unaltered, and keeping machine-generated images distinguishable from photographs — an obligation created by generative image tools rather than inherited from film.
- Real precedent: Leica's M11-P (2023) is the first camera to embed Content Credentials at the point of capture, under the Coalition for Content Provenance and Authenticity (C2PA) standard backed by Adobe, Microsoft, Sony and the BBC.
- Identity-chain continuity: strengthens rather than dissolves the professional and photojournalistic identity chain, since provenance is a capability a camera can hold, and one a phone screenshot fakes only with difficulty.
- Capability match: open to established camera makers with a hardware root of trust, a capability a phone-camera software update rarely confers on its own.
- Ruled out: watermarking after export, which a bad actor can strip; the obligation is only removed if the credential is bound at the sensor rather than added afterward.
- What is expected: standard-issue provenance capture spreading from professional cameras to smartphone camera modules, as the volume of synthetic images makes unverified images a growing, rather than shrinking, obligation.
- Falsifier: adoption tracked across camera-module shipments; the direction fails if credentialed capture stays under 5 percent of new devices three years after a majority of major sensor makers commit to the standard.

- **Candidate 3 — auto-curated, ambient display.**

- Obligation removed: selecting, printing, and mounting a physical photograph for display, and the obligation to update that display as life moves on.
- Real precedent: Aura Frames and Skylight Frame already ship Wi-Fi picture frames that update from a phone, skipping the printing step entirely; Google Photos' and Apple Photos' automatic "Memories" features already perform the selection obligation algorithmically.

- Identity-chain continuity: takes the identity chain of the printed portrait on the wall and re-attaches it to a frame that updates itself, rather than removing display from the home.
- Capability match: open to any company holding cloud photo storage and a connected display, which includes both an incumbent photo company and a new entrant.
- Ruled out: a frame that requires manual selection from a phone app for every update, which re-introduces the obligation the frame was bought to remove.
- What is expected: fully automatic selection, using the same on-device models already used for search, becoming the default rather than an optional mode.
- Falsifier: 50 households, 90 days; failure if manual re-curation happens more than once a week per household.

Books, telephony, drinking water, and home heating: two candidates each

• **Books.**

- *Recall layer over what was read.* Obligation removed: remembering and re-finding a passage read previously, rather than remembering to read at all. Real precedent: Kindle's "X-Ray" and Readwise already extract and resurface highlights automatically. What is expected: a recall layer that surfaces a forgotten passage unprompted, at the moment a later conversation or decision needs it, rather than only on manual search.
- *Narration that skips the obligation to record it.* Obligation removed: paying a narrator's studio time, which currently limits how many backlist titles ever become audiobooks. Real precedent: Audible's and other publishers' AI-narrated audiobook programmes, launched from 2023 onward, already convert backlist text titles using existing text rather than new studio recording. What is expected: the obligation removed is the publisher's cost of narration, not the reader's experience, so the direction favours backlist availability ahead of further gains in narration quality.

• **Telephony.**

- *Notification triage held by an agent, not a screen.* Overlaps with the Apple and OpenAI cards of Section 11; the real precedent is already the same one cited there (on-device summarisation features shipping in 2024–2026 phone operating systems).
- *Presence held apart from a call.* Obligation removed: the scheduling and turn-taking obligation of a live call, for the subset of calls that only convey a status. Real precedent: asynchronous voice-message features already shipping inside messaging apps, used as a scheduling-free substitute for a subset of calls.

• **Drinking water.**

- *A vessel that removes the memory obligation.* Obligation removed: remembering to drink water across a day, rather than the taste or purity obligations already removed by pipes and filters. Real precedent: HidrateSpark's Bluetooth-connected bottles already track intake and prompt by glowing, syncing to phone health apps. What is expected: the prompt moving from the bottle itself to whichever device is already being looked at, since a glowing bottle only works if it is within sight.
- *Provenance for bottled water, on the label the candle lost.* This is a leave-alone candidate. Bottled water sold on origin rather than on purity is the survivor niche of Section 6.1's disappearance test, kept by an identity chain rather than an obligation gap; continuation of the niche is what is expected here, rather than a jump.

• **Home heating.**

- *Room-level rather than whole-house inference.* Obligation removed: the guesswork of a single thermostat standing in for every room's actual comfort. Real precedent: Ecobee's room sensors and Google Nest's multi-sensor integrations already ship room-level occupancy sensing layered onto a single connected thermostat. What is expected: the room, rather than the house, becoming the unit the system reasons about, using only sensors already sold as accessories, ahead of a new device class.
- *Pre-emptive rather than reactive adjustment.* Obligation removed: the lag between a change in occupancy or weather and the system's response. Real precedent: connected thermostats already using weather-API and calendar data to pre-heat or pre-cool ahead of arrival. What is expected: the same pre-emption extended to health and air-quality triggers (pollen forecasts, wildfire smoke advisories), alongside temperature.

Key points.

- Every candidate above is a method-generated hypothesis, in the sense used throughout this paper: a direction the four-variable model favours, not a claim already confirmed.
- Every candidate is anchored to a real, already-shipping precedent rather than a speculative technology, which is what "extreme high quality" forecasting can offer that a purely imaginative scenario cannot: a documented base rate for the capability required.
- Several candidates per domain are given deliberately, because the model's claim is about direction and structure, not about picking a single winner in advance; Section 9's falsifiers apply to each candidate individually.

11. Three companies, deeper

Each forecast below comes from the positioning dissection of Section 12, run on public product records. Each is a method-generated hypothesis; verdicts are ordinal outcomes of comparing candidates against the frontier on the four variables of Section 9; the falsifier is the cheapest test that would refute the jump.

11.1 Versuni

- Position: the home's regulated states — cooked food, coffee, clean floors, wearable garments, breathable air, a safe home — through appliances people already own, under trusted brands [15–20].
- Obligations left in place across the portfolio: consumables, cleaning of the appliance itself, presence at the appliance, scheduling, and interpretation of readings — five clusters, repeated in every category.
- Capabilities: sensing inside the appliance, actuation at the point of use, a connected layer, identity in coffee and kitchen, distribution into nearly every home.
- The jump (Figure 10): the obligation-free loop, appliance by appliance and then across them — the deadline cooker (endpoint sensing; the meal is ready at the stated time), coffee that precedes the wish, the finishing hanger that removes garment care from the ironing episode, the air envelope loop with ventilation partners, safety interlocks — with one shared consumable-and-cleaning service under the brand. The dependency changed is presence and scheduling.
- Directions to go: cross-appliance loops; the hanger as the unit of garment care; the kitchen that cooks by deadline; the air loop; identity in kitchen and coffee, where the chain is already strong.
- Directions to leave alone: presets and modes; a voice assistant as a feature; robot vacuums on mechanism alone against rivals who hold that frontier; a decorative refresh with zero obligation removed; categories with zero sensing advantage.
- Disappearance risk: the stand-alone iron, once the hanger loop removes its need at storage; the single-function kettle and toaster, absorbed into loops; any product whose consumable is its business model once a rival removes the consumable.
- The jump, stated as a forecast: the deadline cooker with the finishing hanger, removing two daily obligations every household still pays. Falsifier: 200 meals, blinded doneness, failure under 85 percent unaided; 50 households, 60 days, failure if ironing episodes drop by less than 50 percent.

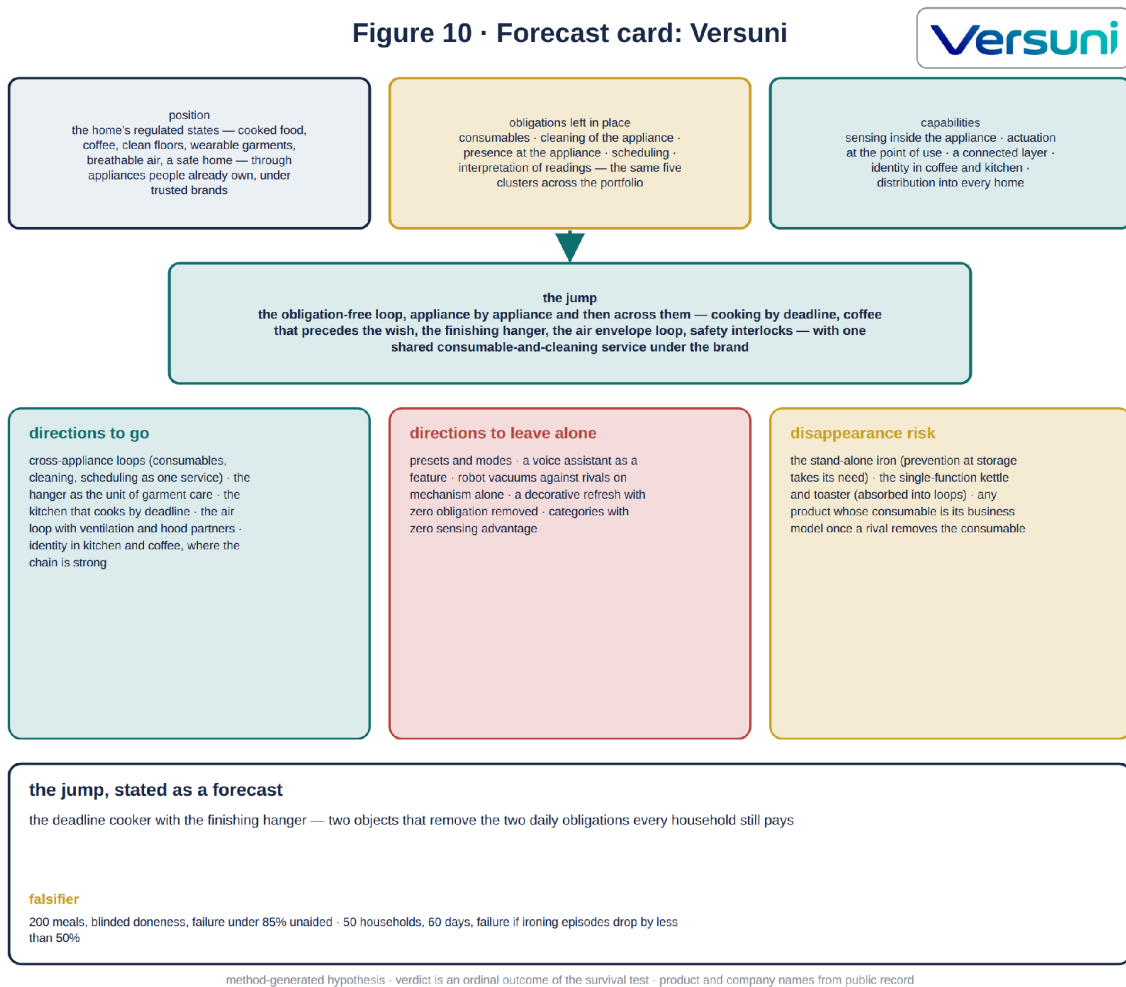


Figure 10. Forecast card, Versuni. Position, obligations, capabilities, the jump, directions to go and to leave alone, disappearance risk, and the falsifier.

11.2 Apple

- **Position:** the pocket computer and the objects on the body — phone, watch, earwear — with the strongest identity chain in consumer technology, trust on privacy, its own silicon, distribution to the wrist and the ear.
- **Obligations left in place:** the phone must be carried, charged daily, taken out and looked at; the screen is the interface for almost everything; attention is taxed by notifications; applications are juggled by hand.
- **Capabilities:** sensing on the body, silicon that runs models on the device, the identity chain, trust, a store that turns every device into a platform.
- **The jump:** the phone that stays in the pocket — the ear-and-eye loop, in which earwear and glasses perceive, a model on the device decides, and the person is spoken to, or shown, only what the desired state requires. The dependencies changed are the screen as interface and the hand as actuator.
- **Directions to go:** hearing and vision as regulated states on the body; health as a loop rather than a dashboard; a device that speaks less and does more; models on the device, where privacy is the bundle.
- **Directions to leave alone:** the headset as a trend, carrying a heavy obligation set for paths the phone already opens; foldable screens, a feature addition to the screen dependency; a chatbot as a feature rather than a loop; a car, where few obligations are removed that incumbents leave in place.

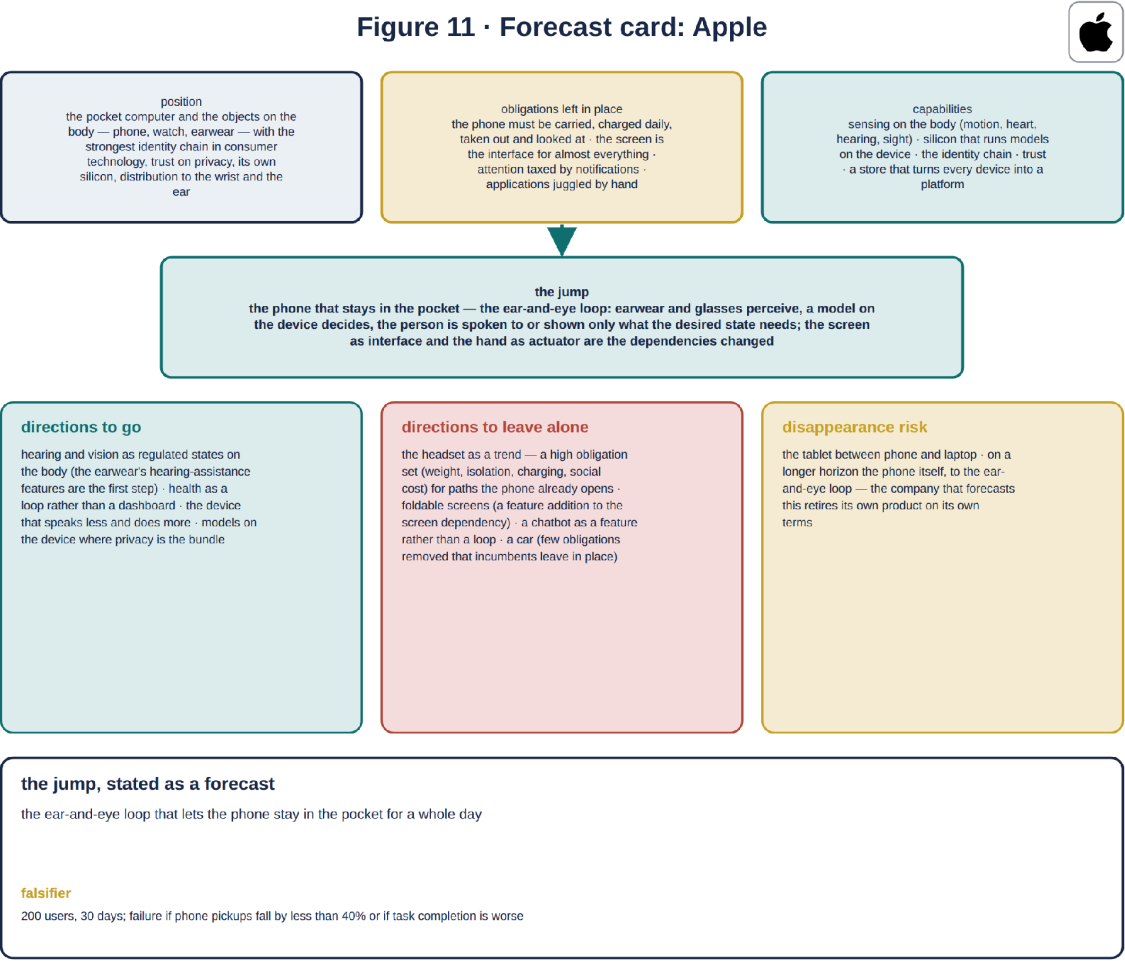
- Disappearance risk: the tablet, positioned between phone and laptop; on a longer horizon, the phone itself, to the ear-and-eye loop.
- The jump, stated as a forecast: the ear-and-eye loop that lets the phone stay in the pocket for a whole day. Falsifier: 200 users, 30 days; failure if phone pickups fall by less than 40 percent or if task completion is worse.

11.3 OpenAI

- Position: the answer engine — a model that writes the answer or the draft, reached through a chat and a developer surface, with a distribution measured in hundreds of millions of people.
- Obligations left in place: the person must ask, supply context each time, verify what comes back, move the output into the world by hand, and attend to a chat window. The answer arrives; the outcome stays the person's responsibility.
- Capabilities: the models, the developer surface, the distribution, a growing ability to act inside tools on the person's behalf.
- The jump: from answering to holding a state — an agent that takes a desired state, perceives the person's world with consent, acts inside tools, verifies its own result, and speaks only when the state is reached or a veto is needed. The dependency changed is the ask-and-verify loop, moved from the person into the machine.
- Directions to go: verification as the product; the perception side, with consent, so the ask disappears; agents that hold one state well; the developer surface as the place other companies build their loops.
- Directions to leave alone: modes and features in the chat box; hardware for its own sake, where a device with zero obligation removed is a carrying cost; benchmark scores pursued as an end; advertising inside answers, a block on the trust dependency the jump depends on.
- Disappearance risk: the chat interface itself, once agents hold states directly; answer-only products in general.
- The jump, stated as a forecast: an agent that stays silent until the thing is done and checked. Falsifier: 100 users, 60 days, three household or work states; failure if states are held less than 80 percent of the time or if unverified errors exceed one per week.

11.4 Reading the three together

- The three jumps share one form: presence, asking and looking are the obligations dissolved; the machine perceives and verifies; the person keeps the veto.
- Each leave-alone list contains a direction the market is currently rewarding. Trends reward the attribute of the present embodiment; the forecast rewards the removal of an obligation, which is a different axis.
- Each disappearance risk names one of the company's own products — the practical form of a subtractive forecast: the company retires its own faster horses on its own terms, in the sense of Section 6.2's counterfactual.



method-generated hypothesis - verdict is an ordinal outcome of the survival test - product and company names from public record

Figure 11. Forecast card, Apple. Same layout.

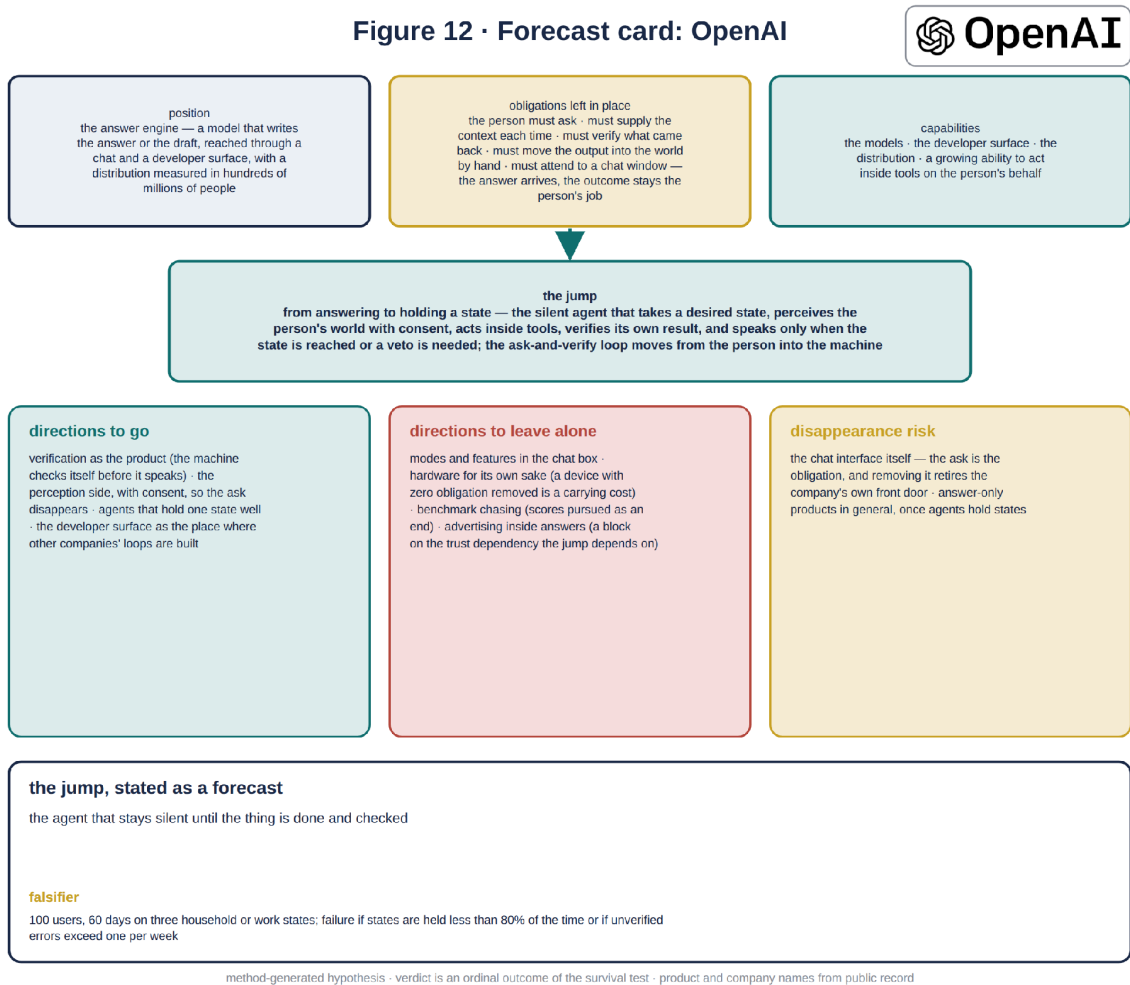
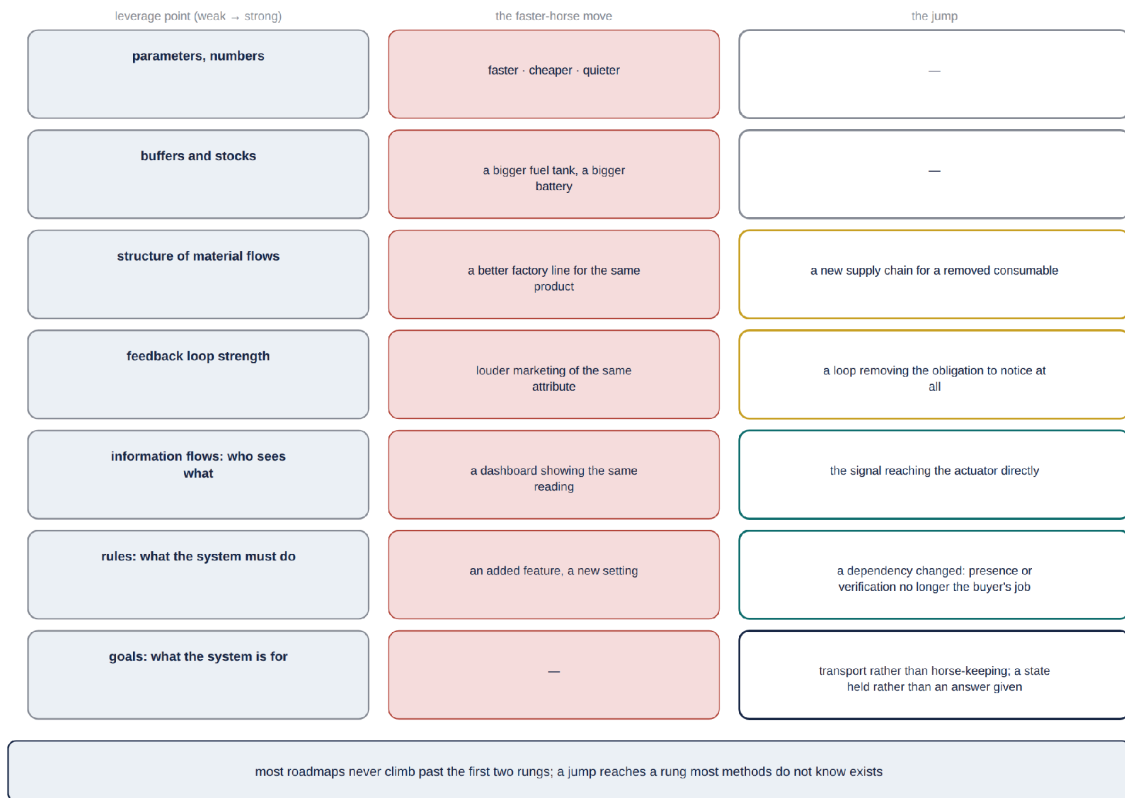


Figure 12. Forecast card, OpenAI. Same layout.

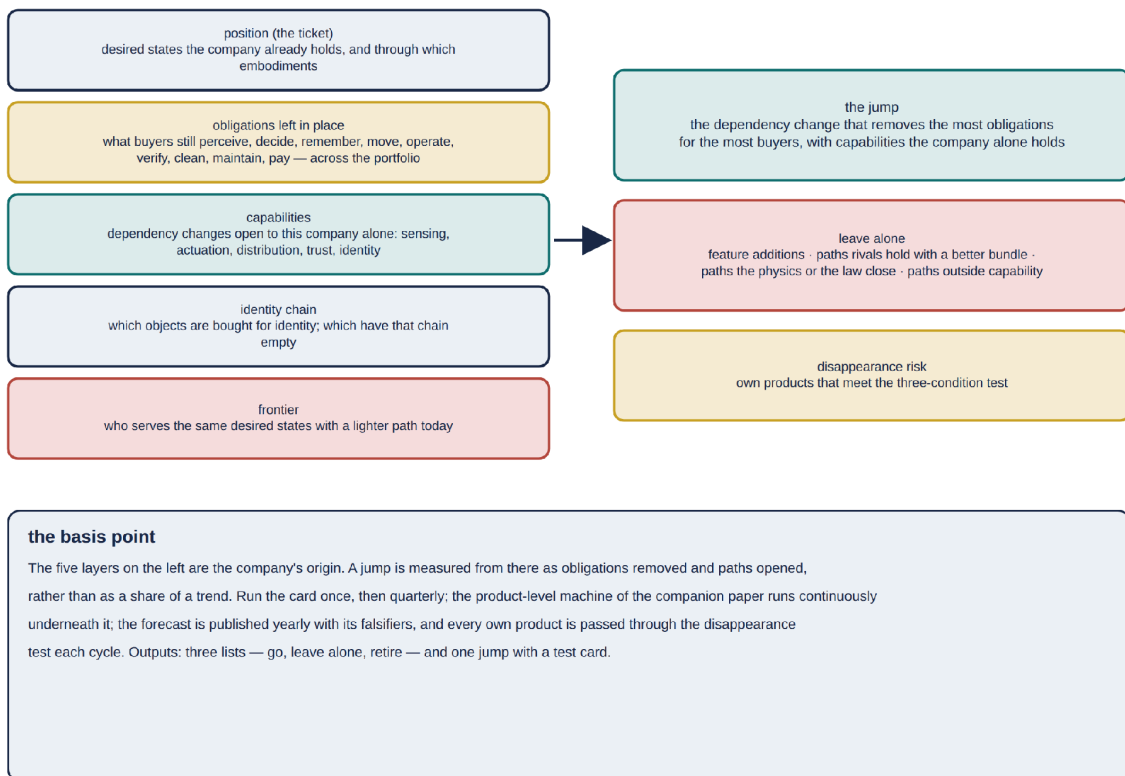
Figure 13 · Meadows' leverage-point ladder, applied to a jump**Figure 13. Meadows' leverage-point ladder, applied to the faster-horse roadmap and to a jump.** Read alongside Table 7.

12. How to apply it to a company: the basis point

A company is dissected with the same structure as a product, at portfolio level (Table 10). The result is the basis point — the origin from which every jump is measured.

Table 10. The positioning dissection.

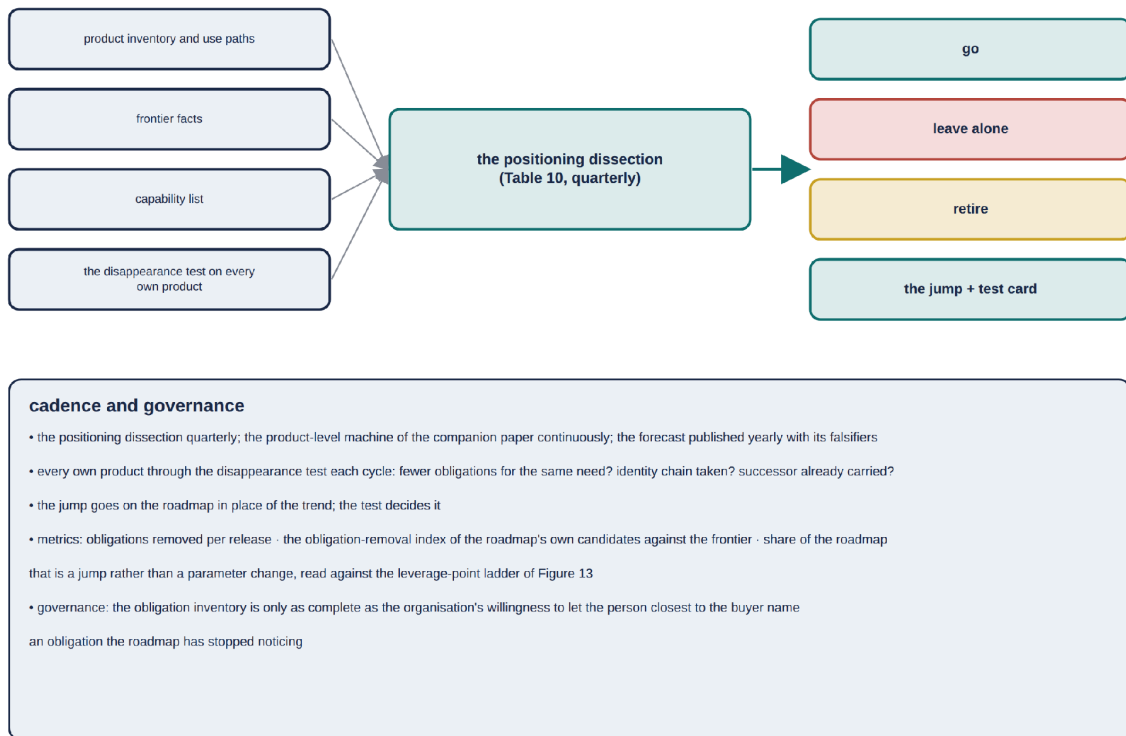
Layer	Question for a company	What it yields
Position (the ticket)	Which desired states does the company already hold for its buyers, through which embodiments?	The list of tickets to play
Obligations left in place	Across the portfolio, what must buyers still perceive, decide, remember, move, operate, verify, clean, maintain and pay?	The obligation inventory — where the jumps hide
Capabilities	Which dependency changes are open to this company alone — sensing, actuation, distribution, trust, identity?	The company's own directions
Identity chain	Which objects are bought for identity, and which have that chain empty?	Lindy assets; disappearance risks
Frontier	Who serves the same desired states with a lighter path today?	The opponents in the survival test
The jump	The dependency change that removes the most obligations for the most buyers with capabilities the company alone holds	The forecast
Leave alone	Feature additions; paths rivals hold with a better bundle; paths physics or law close; paths outside capability	The directions to leave alone
Disappearance risk	Which own products meet the three-condition test against a lighter embodiment?	What to retire or re-embody first

Figure 9 · The positioning dissection: the company card and the basis point**Figure 9. The positioning dissection.** The company card and the basis point from which a jump is measured.

Applying it, in order:

- List every product and every area of work; for each, write the use path and the desired state it serves.
- Write the obligation inventory from the use paths rather than from opinions; rank obligations by how widely they are paid.
- List the capabilities the company alone holds, and the frontier for each desired state.
- Run the disappearance test on every own product.
- Score candidate jumps on the four variables of Section 9; keep the ones with the largest positive obligation-removal index and identity-chain continuity.
- Publish three lists — go, leave alone, retire — and one jump with a test card.

13. How to build the machine

Figure 14 · The forecast machine: inputs, cadence, outputs**Figure 14. The forecast machine.** Inputs, cadence, outputs.

- Inputs: the product inventory and use paths; frontier facts; the capability list; the disappearance test on every own product.
- Core: the positioning dissection, run quarterly; the four-variable scoring of Section 9 applied to every candidate jump under consideration.
- Outputs: go, leave alone, retire, and the jump with its test card. The jump goes on the roadmap in place of the trend; the test decides it.
- Cadence: the product-level innovation machine of the companion paper [1] runs continuously underneath; the forecast is published yearly with its falsifiers and revisited against the tests.
- Metrics: obligations removed per release; the obligation-removal index of the roadmap's own candidates against the frontier; share of the roadmap that is a jump rather than a parameter change, read against Meadows' leverage-point ladder (Table 7).
- Governance: the culture condition of Table 6 applies here directly — an obligation inventory is only as complete as the organisation's willingness to let the person closest to the buyer name an obligation the roadmap has stopped noticing.

14. Limits and falsifiers

- The forecast is ordinal and positional. It ranks directions for a stated company and refuses to date them; timing is left to trend methods, which remain the correct tool for that specific input.
- The historical backtest of Section 9.3 is a counterfactual calibration on a finite, hand-coded set of transitions, weaker than a genuine forward record; the prospective protocol of Section 9.4 exists to supply that record.
- Coding the four variables from historical record stays only partly blind to the outcome, since the outcome is itself part of the history being coded; Section 9.3's pre-registration states this limit and sets the inter-rater agreement threshold that partially controls for it.

- The AFI law, $\text{Freedom}(x) = (\text{Perception}(x) / \text{Distortion}(x))^{\alpha\infty}$, is used here only in its ordinal reading — the lightest-path preorder behind the obligation-removal index — with its negative results kept in view [13, 24]; the readout stays out of selection, which runs on the path preorder alone.
- The paper's claim stands or falls on three conditions: the four-variable model matching or trailing trend extrapolation in the backtest (Section 9.3); a published prospective forecast refuted at its own threshold (Section 9.4); or a documented disappearance satisfying fewer than two of the three conditions in Section 6.1.

15. Conclusion

Pattern recognition reads the frequency of what has already happened; an innovation jump carries zero frequency, being still absent from the buyer's experience. Path recognition reads the structure that is already there — the obligations a buyer carries, the identity chain attached to the present embodiment, the capabilities a specific company holds — and states a direction and a test rather than a magnitude and a date. Kahneman's account of judgment explains why organisations default to the pattern; Taleb's account of fat tails explains why the pattern fails structurally rather than by degree; systems theory reframes the jump as an intervention at a high leverage point rather than a tweak to a parameter; and six other fields, from meteorology to team culture, already forecast this way under conditions of their own. A specific model — four variables, a historical backtest, a prospective protocol — is proposed here for testing, stress-tested across five unrelated domains and stated for three live companies. Every claim in this paper is a method-generated hypothesis, and every jump carries the test that would refute it.

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Glossary

- **Basis point: the company's present position** — tickets held, obligations left in place, capabilities, identity chains, frontier — from which every jump is measured.
- **Capability match**: whether a company already holds the sensing, actuation, distribution, trust or identity capability a candidate jump requires.
- **Desire**: a state the speaker wants, typically phrased inside the present embodiment ("a faster horse").

- **Disappearance test: three conditions a product must satisfy together to be at risk** — fewer obligations for the same need; identity chain taken or made irrelevant; successor already carried for another reason.
- Embodiment: the present mechanism by which a desired state is reached (the horse, the film camera, the chat box).
- Expectation: a narrative that the future resembles the present; judged by coherence rather than by a test.
- Extremistan: Taleb's term for a domain where a single observation can dominate the aggregate outcome; the domain innovation forecasting occupies.
- Forecast (path-recognised): a statement of which obligation dissolves next, by which embodiment, from which position, with a falsifier.
- Fourth quadrant: Taleb's region of complex payoffs under fat tails, where statistical prediction fails structurally; where innovation jumps live.
- Frontier gap: the obligation-removal advantage, or disadvantage, of a candidate jump relative to the best current rival or substitute.
- Identity chain: the reason an object is bought for who its owner is, beyond what it does; the Lindy asset.
- Identity-chain continuity: whether a candidate embodiment takes, dissolves, or leaves intact the identity chain of the incumbent it replaces.
- Jump: a change of embodiment that removes obligations; the object of the forecast.
- Leverage point: a place to intervene in a system, ranked from weak (a parameter) to strong (a rule or a goal); a jump intervenes at a strong leverage point.
- Lindy effect: the expected remaining life of a technology grows with its age; applies to objects with an identity or ritual chain.
- Mediocristan: Taleb's term for a domain where any single observation moves the aggregate by very little; classical statistics applies safely there.
- Obligation: what a buyer must perceive, decide, remember, move, operate, verify, clean, maintain or pay to reach a desired state through the present embodiment.
- Obligation-removal index: the count of obligations a candidate embodiment removes relative to the incumbent, for the same primary need.
- Passible: reachable from the present state by this agent along an executable path.
- Path recognition: reading the obligation, identity and capability structure that holds an embodiment in place, as the basis for a forecast.
- Pattern recognition: reading the frequency of a remembered series and projecting it forward.
- Position: the desired states a company already holds for its buyers, and through which embodiments.
- Prediction: a statement that an event occurs, with a probability, scored by calibration.
- Retire: the verdict on an own product that meets the disappearance test.
- **Test card: the falsifier attached to a jump** — population, duration, threshold.
- Trend: the slope of an attribute of the present embodiment; an input to what becomes passible; direction is supplied by path recognition, not by the trend.